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Life-cycle greenhouse gas emissions and net energy assessment of large-scale hydrogen production *via* electrolysis and solar PV†

Graham Palmer, *[a] Ashley Roberts,[b] Andrew Hoadley,[c] Roger Dargaville[d] and Damon Honnery[a]

Water electrolysis powered by solar photovoltaics (PV) is one of several promising green hydrogen production technologies. It is critical that the life cycle environmental impacts and net energy balance are assessed to ensure that solar-electrolysis can contribute to the deep decarbonisation of global energy. Life cycle assessment (LCA) and net energy analysis (NEA) are tools for environmental and net energy assessment of such technologies. LCA/NEA studies of renewable hydrogen typically include simplifying assumptions, such as steady state operation under average conditions. Whilst simplifications may be necessary for preliminary analysis, marked differences arising from context specific variances and operating constraints may be overlooked. To address this gap, we conduct an LCA/NEA of a hypothetical large-scale solar-electrolysis plant, with a focus on operational sensitivities. We find the most significant component is the solar modules due to the materials and processes used in their manufacture. We find the most significant sensitivity stems from the electrolyser turndown and the commensurate need to buffer solar electricity with storage or grid electricity. Under baseline conditions, the greenhouse gas (GHG) emissions are around one-quarter that of the currently dominant process for hydrogen production, steam methane reforming (SMR). However, sensitivity analysis shows that GHG emissions may be comparable to SMR under reasonably anticipated conditions. Net energy results are less than for fossil fuels and sufficiently uncertain to warrant further attention. We recommend that LCA and NEA are integrated with project planning to ensure that hydrogen meets the goals of green production.

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Broader context

Hydrogen is expected to make a significant contribution to the deep decarbonisation of energy used in transport and industry. Several hydrogen production pathways are potentially low carbon, with much of the effort being directed at hydrogen production *via* renewable energy and water electrolysis. A significant barrier to large scale implementation is high cost, with most attention given to reducing the capital cost of renewable and electrolyser plant and equipment. However, the degree to which renewable hydrogen can contribute to deep decarbonisation depends on several factors that are only partly related to cost. These include the strength and variability of the solar or wind resource, real-world operating performance of renewable-electrolysis plants, the energy and greenhouse gas emissions embodied in the global supply chain, and the availability of critical minerals used to construct plants. In principle, renewable-electrolysis offers a sustainable pathway towards deep decarbonisation. A further factor is that the net energy and environmental impacts may be worse than expected under reasonably anticipated conditions. While small-scale facilities have provided valuable information in the early development of renewable-hydrogen, and commensurate cost declines, there is a need for greater understanding of the operation and performance of plants at the scale needed to contribute to deep decarbonisation.

1. Introduction

Hydrogen is expected to play a significant role in the deep decarbonisation of transport and industry, especially for energy services that will be difficult or expensive to electrify. Hydrogen can be produced from a diverse range of primary energy resources, including renewable and nuclear energy, biomass,

[a] *Department of Mechanical and Aerospace Engineering, Monash University, Clayton, Victoria, Australia. E-mail: graham.palmer@monash.edu*
[b] *Woodside Energy, Perth, Western Australia, Australia*
[c] *Chemical Engineering, Monash University, Clayton, Victoria, Australia*
[d] *Department of Civil Engineering, Monash University, Clayton, Victoria, Australia*

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natural gas, coal, and oil.[1] The 'clean' and 'low-carbon' labels can be applied to several production pathways,[2,3] however, the 'green' label is generally reserved for hydrogen produced almost entirely with renewables. Much of the international policy discussion has focused on green hydrogen deployment, with an increasing number of jurisdictions announcing policies to accelerate deployment.[2] Water electrolysis powered by solar photovoltaics (PV) is one of several promising candidates, but requires further capital cost reductions to make it competitive with other technologies.[4,5]

The dominant process for hydrogen production is currently steam methane reforming (SMR) of natural gas,[2] which we treat as the reference technology. SMR plants operate continuously with an operating rate of typically greater than 80%.[6] SMR has a relatively high hydrogen-to-carbon ratio, which minimizes the production of carbon dioxide.[7] The greenhouse gas (GHG) emission intensity is 8.9 to 11.9 kg CO_2 e kg^{-1} H_2.[6,8] Most of the emissions arise during operation and can be determined by measuring the flow and composition of the natural gas entering the system as reformer feed and combustion fuel. The flow proportions are 60–70% feed, and 30–40% fuel.[9] The reform process produces a concentrated stream of CO_2, which is readily captured, while combustion produces a dilute stream, which is more expensive and energy intensive to capture. It may also be feasible to electrify heat and steam generation,[10] and thereby reduce GHG emissions by sourcing electricity from renewable sources.

In contrast to SMR, electrolysis powered by solar PV produces almost no direct emissions during operation, however the infrastructure is energy and materials intensive. To the extent that fossil fuels are used to manufacture, construct, and operate the infrastructure, the impacts of extracting and burning those fossil fuels will be indirectly embedded in the hydrogen. A global transition from fossil fuels will of course reduce those embedded impacts.

The European 'CertifHy'[11] scheme recognises 'low-carbon' as at least a 60% reduction in greenhouse emissions relative to SMR, or lower than 4.5 kg CO_2 e kg^{-1} H_2. However, the CertifHy methodology does not encompass the indirect emissions, energy or material flows associated with the manufacture and construction of renewable-electrolysis plants.

Net energy analysis (NEA) and Life cycle analysis (LCA) are tools for evaluating the energetic and environmental performance of energy supply technologies. These tools can be used to estimate the direct, and the indirect, energy, material and waste flows associated with hydrogen production. A key metric is the Energy Return on Investment (EROI). EROI is dimensionless, and defined as the ratio of the energy returned from an energy supply process compared to the energy invested in that process.[12] It is conceptually an energy profit ratio and based on the fundamental principle that any supply system must return more energy than has been diverted from other uses to get that energy. Resource depletion tends to lower EROI while technological improvement tends to increase EROI. As a physical metric, EROI can be useful for identifying physical constraints that may not be obvious from techno-economic analyses, and the associated impacts of taxes, subsidies, exchange rates, and price dynamics.

Despite technological improvements in oil and gas extraction, renewables, and smart grids, there has been a marked decline in the EROI of the global energy system in recent decades, leading to concerns that a further decline may impede energy transition.[13,14] Two recent studies of global energy find a final stage EROI of 6.0 and 8.5 respectively.[15,16] Given that fossil energy sources make up 84% of global commercial primary energy supply,[17] the global estimates are a proxy for current fossil fuel EROI. Since these studies encompass direct and indirect energy using an input–output methodology, they are representative of the system boundary of this work and provide a reference point for comparison.

The goal of this study is to assess the life cycle net energy balance and greenhouse gas (GHG) emission performance of large-scale water electrolysis powered by solar PV in a high insolation region with a focus on operational sensitivities. The context is the production of hydrogen for industrial applications, downstream processing, or for export. The system boundary encompasses the direct flows that are the focus of hydrogen certification schemes, as well as the indirect flows that comprise most of the impacts associated with solar-electrolysis. Two baseline scenarios are considered – a solar-battery scenario without grid connection; and a solar-grid scenario that uses grid balancing of variable solar supply.

1.1 The magnitude of the upscaling required

An important consideration of solar-electrolysis in the context of climate mitigation is the enormity of upscaling required—both at the global scale with respect to the investment, land area, materials, and embodied energy; and at the project scale with respect to the potential localised impacts of gigawatt scale plants. According to the IEA,[2] less than 0.1% of hydrogen is currently produced *via* water electrolysis and only a fraction of this production is powered by renewable energy.

Taking IRENA's REmap scenario as a reference, renewable hydrogen could deliver 5% of total final energy demand in 2050.[18] Assuming that a half of this demand is met by solar-electrolysis, 3100 GW of solar would need to be dedicated to hydrogen production in 2050, or around four-times the current world solar PV installed capacity (based on the current world capacity factor for solar PV of 14%[19]). These projections would imply that perhaps hundreds of gigawatt-scale plants will need to be in operation by 2050. Targets for hydrogen demand beyond 2050 are much greater.[20]

Along with hydrogen production and use, an energy transition will involve the synchronous upscaling of renewable electricity, batteries, and energy use technologies.[21] The materials and metals demanded by a low-carbon economy are projected to be immense and create challenges along the full supply pathway.[22] Some of the most critical materials include cobalt, lithium, nickel, indium, silver, and tellurium.[23] For hydrogen, the platinum group metals used in PEM electrolysers and fuel cells are critical.[24] There is no immediate concern for copper resources, but the average ore grade is declining as higher grade deposits become exhausted.[25] Energy extraction and processing costs increase super-linearly with declining ore grade, and therefore will tend to worsen EROI.[26]

In light of the sheer scale of the hydrogen challenge, several questions demand close consideration. For instance, what will the costs be for water electrolysis powered by solar PV, in energy and material terms; what are the trade-offs between hydrogen and electric transition pathways; and how will energetic, financial, social and other constraints determine or shape future hydrogen production pathways?[27]

1.2 Water electrolysis powered by solar PV

While more advanced, integrated solar-electrolysis systems are under development, the current design of a solar-electrolysis plant for hydrogen production will contain separate modules for the solar PV and water electrolysis. Solar PV and water electrolysis are separately mature technologies, however there is limited operating experience with these technologies as large-scale integrated systems. Solar-electrolysis presents operational challenges that are different from the challenges associated with grid connected solar electricity. One challenge is determining the optimal strategy to manage the natural variability of solar generation. This will require decisions around technologies, vendors, and context. Of the three main electrolyser technologies, proton exchange membrane (PEM) electrolysers possess the highest operational flexibility and turndown capability.[28] Turndown expresses the reduction in operating capacity of a plant from its rated capacity, and is expressed as a percentage reduction (*i.e.* 90% turndown refers to operation at 10% of rated capacity). However, gigawatt-scale PEM electrolyser plants have yet to be constructed, and post-electrolysis processing of H_2 remains a challenge. For example, the dynamic response and turndown capability of ammonia plants and hydrogen liquefaction plants is much less than for PEM electrolysis.[29] Alkaline electrolysers (AE) have been adopted for this study because they are the incumbent technology for industrial applications, have a higher market share, lower capital cost, and superior durability.[28,30] Furthermore, the materials in AE are less exposed than PEM electrolysers to long term supply risks.[31] Solid oxide electrolysis cells (SOECs) are the least developed of the three main electrolyser technologies.[2]

Water electrolysis, hydrogen compression, liquefaction, and ammonia production, are subject to physical and thermodynamic operating constraints. The high thermal inertia and high operating pressures of some processes reduce permissible ramp rates, and preclude frequent start-ups and shut-downs. It may be possible for plants to be operated outside of their design envelope, but doing so may impact controllability and possibly safety, and may accelerate degradation of plant components. Chemical plants are ordinarily optimised for steady-state operation at low to medium turndown.

The challenges of powering electrolysers with a variable power source have been identified in the research literature,[32] hydrogen economic modelling literature[33] and commercial domain,[34] but not adequately incorporated into the environmental assessment literature. Operational dynamics are identified in some LCA studies, but are not included in detailed modelling. For instance, Yadav and Banerjee[35] identified the need for PV buffering and included batteries but did not undertake a detailed assessment of operational dynamics. Instead, the LCA was conducted with steady state flows.

An alternative strategy to buffering the electricity pre-electrolysis is buffering the hydrogen post-electrolysis. Mallapragada *et al.*[33] found that pressurised hydrogen storage was far more cost effective than battery storage of electricity on a per kg H_2 basis, although Mallapragada *et al.* assumed fully flexible operation of the electrolyser and compressor, and 100% electrolyser turndown for a solar PV powered system.

1.3 Future of grid emission intensity

To date, the principal strategy for managing renewable variability of pilot-scale plants has been grid buffering. However, grid buffering may significantly increase GHG emissions, even where grid imports comprise a minor share of energy.[36] The 'renewable' electrolysis plants that are currently operating are typically powered by renewable power indirectly *via* market instruments such as power purchase agreements (PPAs). The impossibility of tracing electron flows leads to uncertainty as to the source of electricity unless there is a direct physical connection between generator and load.[37] Even if an electrolysis plant is located adjacent to a wind or solar farm, as long as both are tied to the grid, electrolysis will embed the emissions of the grid, rather than the (zero direct) emissions of the adjacent renewable farm. Two examples are the Energiepark Mainz wind-electrolysis plant in Germany,[38] and the HyBalance wind-electrolysis plant in Denmark.[39] Both facilities adopt PEM electrolysers because of their fast, dynamic response; both are grid connected; and both provide grid support services such as demand response.

The GHG intensity for grid powered electrolysis is significantly greater than for SMR except where grids have a very low emissions factor.[40] Bareiß *et al.*[36] focused on the potential to limit electrolyser use to only those periods of high instantaneous renewable penetration. With the 2017 electricity mix in Germany, the GHG emission intensity would have been 29.5 kg CO_2 e kg^{-1} H_2, which is around 3-times that of SMR. They found that the GHG emission intensity could potentially be reduced to 3.3 kg CO_2 e kg^{-1} H_2 in a future German grid by restricting electrolyser use to only those periods of 100% instantaneous renewable penetration.

Many of the studies that investigate grid powered electrolysis are market-based assessments. These studies focus on markets and prices rather than on environmental impacts. This work is an investigation of the physical flows of such plants, rather than on the role of markets.

1.4 Renewable hydrogen life cycle assessment

LCA studies find that electrolysis powered by solar PV is less GHG emission intensive than SMR, but higher than wind powered-electrolysis.[40–42] Parkinson *et al.*[43] reports a literature range for solar PV-electrolysis of 2.0 to 7.1 kg CO_2 e kg^{-1} H_2

with a central estimate of 4.5 kg CO_2 e kg^{-1} H_2. These results are based on location-dependant annual average insolation, and steady state operation, and do not include emissions attributable to grid imports. Most studies adopt an annual average hydrogen output, rather than assessing hydrogen output at a shorter time resolution, such as a daily or weekly time period that might be used for production planning. Annual resolution is usually adequate for LCA/NEA studies provided that detailed historical data is available to extract important metrics such as maxima, minima, variances, and seasonal behaviour.

There are few studies that consider the implications of large-scale electrolysis. One such study examined solar hydrogen production *via* photoelectrochemical cells (PEC).[44] The hypothetical plant was rated at 610 tonne H_2 per day facility, equivalent to 1 GW continuous operation. The EROI was 1.7, and the most important sensitivities were solar-to-hydrogen conversion efficiency and operational life of the PEC units.

A dilemma for LCA practitioners is that there are no large-scale renewable-electrolysis plants in operation from which to collect operating data and plant specifications. Traditionally, LCA studies are retrospective studies of existing technologies. Common LCA practice is to compile life cycle inventories using commercial information, such as materials and fuel purchase histories. Much of this data are confidential and accessed through research partnerships or agreements. Other data can be estimated by taking information that is known and applying known principles. For example, the embodied energy of a machined component can be estimated from knowledge of the materials and the type of machining process. There is considerable data on components, but little data on how large-scale integrated plants will actually operate.

Assessment of products and processes that are new or emerging is not a new challenge for LCA practitioners. One method for assessing such products or processes is the prospective LCA methodology.[45] Prospective LCA brings together technology learning curves and scenario modelling so that 'prospective' foreground and background data sets can be calculated.[46] For example, electrolyser learning curves can be extrapolated into the future, and by applying demand scenarios, estimates of efficiency and embodied energy in the future can be derived. A weakness of this approach is that technology extrapolation does not provide operational data from which to draw detailed analysis of plant operation.

LCA and NEA are life-cycle metrics that consider impacts over a lifetime. However, the time phasing of flows can be important for renewable infrastructure because of a significant 'bulk investment' of energy and material at the beginning of the project life cycle, with a 'regular return' of energy throughout an operational life.[47] Unlike the concept of discounted cash flows in financial analysis, there is no equivalent method of discounting energy, material and waste flows against time. The need for a 'bulk investment' at the start of solar-electrolysis projects may lead to delayed energy and climate benefits in the event of a rapid deployment of solar-electrolysis.[48]

Estimates of EROI for solar PV are often contentious and vary between technologies of the same type. Koppelaar[49] found a mean EROI of 9.0 with interquartile values of 6.0 and 12.5 for polysilicon, and a mean EROI of 7.5 with interquartile values of 4.5 and 11.0 for monosilicon. Most of the divergence between studies can be accounted for by methodology, the treatment of intermittency, assumptions about real-world performance, and the choice of primary energy accounting.[50] Studies that retrospectively assess EROI with actual operating data sometimes find the EROI to be markedly lower than assessments determined with modelled data.[50,51] Since there is no universal convention for determining the primary energy equivalence of non-thermal renewables, studies adopt different primary energy conversion factors. This leads to a substantial divergence of EROI for notionally similar studies.

1.5 Facility overview

In order to focus on large-scale production for industrial use, the constraints for a hypothetical facility were set as follows: the location was limited to regions with an annual insolation above 2000 kWh m^{-2}; proximity to the ocean or a large water body for electrolysis feedwater; proximity to an industrial centre for infrastructure and support during construction and operation; and wide expanse of land available for solar farms. Regions that conform to these constraints include several countries in the Middle East, the central Pacific Coast of South America, Southern Africa, and several regions in the north and west of Australia. In order to model time-series solar insolation, a coastal location in Western Australia with access to ports and industrial infrastructure was selected.

1.6 Article structure

In this paper, Section 2 discusses methodology, facility overview, description of scenarios, modelling of solar, electrolyser and battery capacities. A detailed system description follows in Section 2.5. Section 3 presents results, sensitivities, and discussion. Section 4 concludes the paper.

2. Methods

2.1 Modelling approach

The LCA modelling approach is process-based, attributional, based on ISO standards 14 040 for LCA. LCA is a method for quantifying cradle-to-grave environmental and net-energy impacts of energy supply systems. The functional unit is one kilogram of dry hydrogen at 80 bar, supplied at the plant gate. A cradle-to-gate boundary is adopted. The production scenario is based on a hypothetical facility producing 36 500 tonnes H_2 per year (average 100 tonnes H_2 per day = 1.1 million N m^3 per day). A high-level, non-optimised design based on heuristic methods was undertaken. System-wide energy and material flows associated with construction, and utilisation of the system were determined. The system boundary includes manufacture, transport, installation and operation of the facility (see Fig. 1).

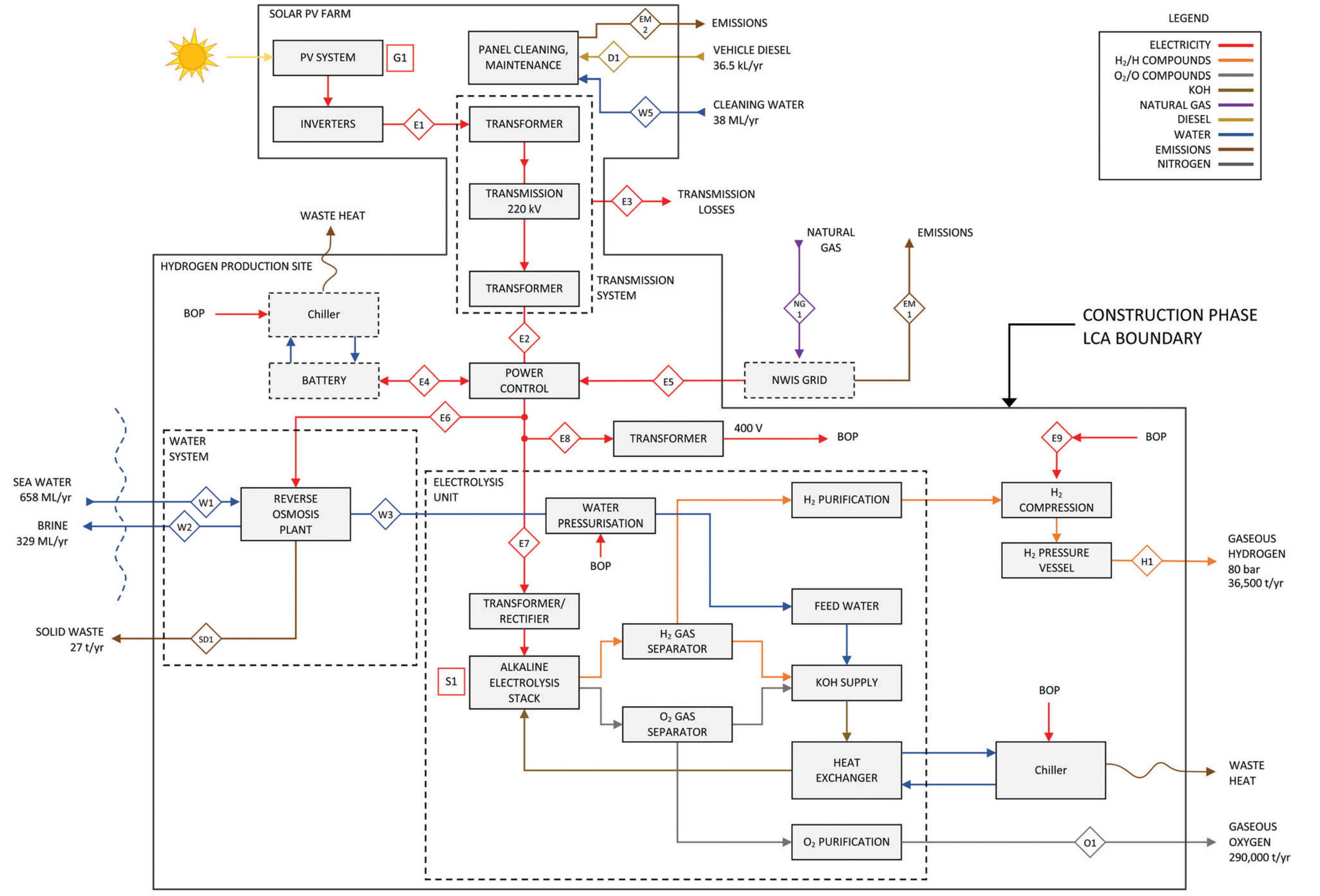

Fig. 1 Process flow diagram and construction phase boundary.

Time-series insolation data were used to capture diurnal and seasonal dynamics. Mass and energy flows were determined based on fundamental principles. All background processes (*i.e.* aluminium, glass, steel, *etc.*) were modelled with ecoinvent version 3.5 LCI database. Chinese production of solar modules was assumed, using the national electricity grid emission intensity from Li *et al.*[52] A sensitivity was conducted for world average electricity grid emission intensity from Ang and Su[53] and projected world average for the IEA Sustainable Development Scenario for 2030[54] (see Table S10, ESI†). The greenhouse gas emissions per unit of hydrogen were calculated as the ratio of the accumulated life cycle greenhouse gas emissions to lifetime hydrogen production. An operational lifetime of 20 years was assumed. For the sensitivity, a lifetime of 15 to 30 years was adopted. Decommissioning was excluded because recycling would likely produce a net environmental benefit.[55] Some components, such as solar frames and hardware, have a lifetime longer than the operating plant, and could be re-used or materials recycled. Based on these findings, the cradle-to-gate indicators would likely be no worse than the cradle-to-grave indicators. Nonetheless, this is a topic for further investigation.

Based on the life cycle inventory energy flows, two net-energy metrics were calculated. The Energy Return on (energy) Investment (EROI) is the ratio of the hydrogen produced to the life cycle energy investments, including construction, maintenance, operation, and decommissioning of the plant, where energy is measured in joules—see eqn (1).

$$\mathrm{EROI} = \frac{E_{\text{supplied}}}{E_{\text{construction}} + E_{\text{maintenance}} + E_{\text{operation}} + E_{\text{decommissioning}}} \tag{1}$$

A related metric, the energy payback time (EPBT), is the ratio of energy investments to the hydrogen energy supplied per year—see eqn (2). It establishes how long a project takes, in years, to return the energy that was initially invested in its construction.

$$\mathrm{EPBT} = \frac{E_{\text{construction}}}{E_{\text{supplied(yr)}}} \quad \text{(years)} \tag{2}$$

The GHG emissions results are reported with respect to the mass of hydrogen. The EROI results are reported with respect to the energy content of hydrogen, using the higher heating value (HHV) of H_2 of 142 MJ kg^{-1}.

2.2 Facility description

The hypothetical production facility consists of two sites—a hydrogen production site and a separately located solar PV farm. The hydrogen production site contains a water desalination plant, electrolysis plant, battery storage system (solar-

Table 1 Facility specifications

Parameter	Description
H_2 production	36 500 tonnes H_2 per year (average 100 tonnes H_2 per day = 1.1 million N m^3 per day)
Functional unit	1 kg dry hydrogen, including dewatering, purification, and compression. Purity is specified as 99.9 vol% with a pressure of 80 bar after compression.
Electrolyser type	Alkaline electrolyser (AE)
Solar farm location	Pilbara coast, Western Australia
Solar tracking	Single axis tracking, north–south orientation
Battery storage	Lithium-ion
Electricity transmission	Dedicated high voltage (275 kV AC) transmission line connects the solar farm with the hydrogen production facility.
Feedwater	Feedwater is supplied *via* desalination. Seawater supply and brine discharge is *via* Hamersley Channel or adjacent King Bay.

battery scenario only) compressor system, and hydrogen storage. The hypothetical hydrogen site is situated near the coast for ready access to feedwater and brine discharge, while the much larger solar farm is located inland. The solar farm occupies a much greater land area than the hydrogen site and need not be located close to a large water body. A distance of 20 km between sites has been assumed, with sensitivity for 100 km and 300 km. The facility specifications are given in Table 1.

A simplified model showing main process flows is shown in Fig. 2 and a more detailed process flow diagram is shown in Fig. 1. As can be seen in Fig. 2, electricity drives two main processes—(a) water desalination from seawater, (b) and water electrolysis. Main waste flows include oxygen, brine and solid wastes. Fig. 1 provides more detail on the flows. The construction phase LCA boundary shows all components considered in the following sections. The electricity grid was outside the construction boundary, however the natural gas use and GHG emissions were included in the operational phase.

2.3 Scenarios

Two scenarios are adopted that are termed 'solar-battery and 'solar-grid'.

(a) For the 'solar-battery' scenario, it is assumed that on-site battery storage powers standby operation with no grid connection, and depending on electrolyser turndown, maintains minimum electrolyser operation during periods of low solar electricity.

(b) For the 'solar-grid' scenario, it is assumed that the hydrogen site is connected to the regional grid – the North West Interconnected System (NWIS) of Western Australia. The grid connection enables grid imports to support load balancing of variable solar supply; and grid exports during surplus solar generation (*i.e.* solar generation exceeding electrolyser capacity). The NWIS comprises mostly gas-fired generation and has a GHG emission factor of 620 g CO_2 e kWh^{-1}.[56] This is higher than the global average of 510 g CO_2 e kWh^{-1}.[57]

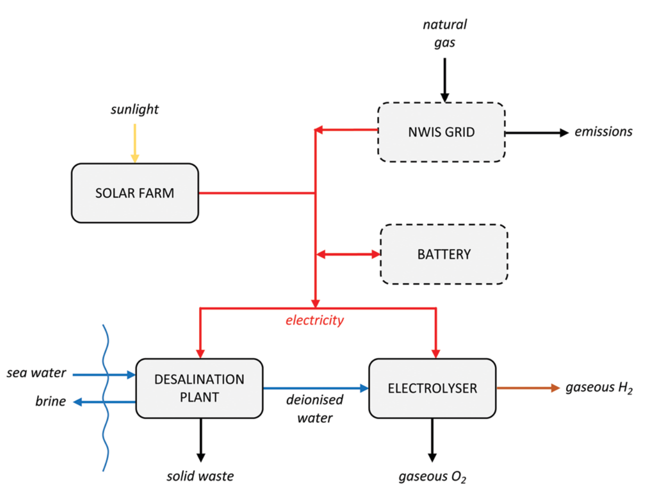

Fig. 2 Simplified model of main process flows.

2.4 Modelling of solar and electrolyser capacities

Hydrogen production was calculated as the ratio of annual solar farm electricity output (minus transmission losses and balance-of-plant loads), and electrolyser efficiency. Solar farm electricity generation was determined using average global horizontal irradiance (GHI) for Learmonth, Western Australia of 2200 kWh m^{-2} $year^{-1}$. Electrolyser efficiency of 55 kWh kg^{-1} H_2 was assumed, with sensitivity of 50 and 60 kWh kg^{-1} H_2. Hydrogen production was subsequently adjusted by the following factors:

(1) Depending on latitude and orientation, solar tracking increases annual solar capture compared to a fixed-tilted plane. A scaling factor of 1.2 was applied to annual generation as being representative of operating Australian solar farms, located in mid latitudes, with north–south single axis tracking.[58]

(2) Component degradation, electrical losses, panel soiling, break-downs and other factors will reduce solar generation. For the purposes of solar LCA, these losses can be aggregated into a single representative 'performance ratio' (PR), of which 0.8 is a typical estimate.[59]

(3) Transmission losses are assumed to be 5.9% for the baseline scenarios. Losses are composed of transformer losses of 2% each and line losses of 2% (see Section 2.5.5).

In order to assess sensitivities at a shorter than annual time resolution, a production model was used. The model was used to determine battery capacity for the solar-battery scenario, and grid flows for the solar-grid scenario. H_2 production was calculated for every minute for one year, corresponding to 1 minute global horizontal irradiance (GHI) data from the

Australian Bureau of Meteorology (BOM). Model parameters were assigned the values given in ESI,† Table S1, and the model decision rules given in ESI,† Table S2. Several years were tested to assess variance between years. For any given annual hydrogen production target, there were many configurations that would satisfy the production target.

For the solar-battery scenario, battery capacity was determined by solving for the lowest battery capacity to meet electrolyser demand for every one-minute period for the year. From a modelling perspective, the only difference between the two scenarios was the constraint that the battery must have previously been charged from surplus solar to enable subsequent discharge, whereas grid imports were independent of grid exports. It was assumed that the battery was fully charged at the beginning of the model run.

Given insolation variability during daytime, and zero insolation at night-time, electrolyser turndown and shutdown/startup specifications are key operating variables. The academic literature reports a maximum turndown for AE of 60 to 90% (lower operating range 10 to 40%).[28,60] However, some commercial vendors report that units can be placed into a standby condition to enable a rapid warm start (*e.g.* GreenHydrogen[61]). NREL reports that AE can be operated with part load, however issues related to safety and control (*e.g.* system pressure, gas crossover) need to be better understood.[32] Much of the testing has been conducted on small-scale units (<100 kW), and there is a need to understand behaviour and degradation modes of large-scale units (>5 MW). There is a trend towards larger units to achieve economies of scale.[62]

In a large-scale facility, electrolysers would likely be installed as modular components, enabling units to be switched on or off depending on the season and prevailing irradiance. During winter, some electrolysers may be placed into an extended cold condition. Other electrolysers may be placed into a standby condition to remain ready for a warm start, and these would draw a small amount of electricity overnight without producing hydrogen. A characteristic of solar PV installed in high-insolation regions is that some electricity generation can be relied upon during the daylight hours, even during extended cloudy periods. For this study, it is assumed that a large-scale facility would possess sufficient flexibility to achieve a system-wide turndown of 95%. Sensitivity analysis is conducted for 80% and 90% turndown. The resulting solar and electrolyser specifications are given in Table 2.

2.5 System description

2.5.1 Solar farm. The solar PV farm comprises 340 watt solar modules, with efficiency of 16.8%,[63] that are mounted in groups of 80 on a single axis assembly. Tracking is single axis, with north–south orientation of the rotating axis. The specifications are given in ESI,† Table S3.

Life cycle inventory (LCI) data for the PV modules are from Frischknecht *et al.*[64] Solar cells were assumed to be multi-crystalline, which until recently comprised the majority of global production. Modules were assumed to be produced in China, which currently accounts for 66% of global production.[63] LCI data for balance of plant (support structure frames, foundations, hardware, wiring, and other miscellaneous components) is derived from Mason *et al.*[65] Inverter LCI is taken from ecoinvent 3.5.

Solar capture is inversely related to panel soiling. Depending on the type of soiling, rain will remove most of it, with some residue remaining.[66] Annual, or more frequent, cleaning is usually carried out on large solar farms.[67] Tractor mounted wet cleaning systems are available that are capable of cleaning 1500 modules per hour. It is assumed that the tractor uses 100 litres of diesel per day. Replacing the diesel with on-site produced hydrogen may be possible, although it is worth noting the reductions in GHG emissions would be negligible because lifetime diesel contributed around 0.05% of the overall lifecycle GHG emissions.

Solar modules and balance-of-plant are assumed to be freighted from Port of Guangzhou China to Port of Dampier Australia, then by road transport to site. Oceanic fuel usage is derived from Notteboom and Cariou.[68] Inverters are assumed to be replaced at 10 year intervals.

2.5.2 Battery system. The battery system comprises lithium ion (Li-ion) batteries, which are the dominant technology for

Table 2 Solar and electrolyser specifications

Parameter		Scenario: Solar-battery	Scenario: Solar-grid
Solar capacity (MW)	[P_{pv}]	1010	940
Solar tracking scaling factor	[S]	1.2	1.2
Transmission losses (%)	[TR]	5.9	5.9
Electrolyser capacity (MW)		550	600
Electrolyser maximum turn down (%)		95	95
Electrolyser capacity factor		0.30	0.27
Battery storage capacity (MWh)		1500	—
Battery power (MW)		375	—
Insolation (kWh m^{-2} year^{-1})	[I]	2200	2200
Performance ratio for solar generation	[PR]	0.8	0.8
Lifetime (years)	[T]	20	20
Lifetime PV electricity delivered to hydrogen site (GW h) $P_{pv} \times I \times \mathrm{PR} \times T \times S \times (1 - \mathrm{TR})$		39 765	36 915
Solar proportion of electricity use		100%	93%
Grid proportion of electricity		—	7%

recent stationary deployments.[69] Nickel manganese cobalt (NMC) chemistry was selected as representative for this application. The specifications for the battery system are given in ESI,† Table S4. Round trip efficiency is 90% based on reported values of 85–95%.[70] Design storage capacity is scaled by 1.2 to account for lifetime capacity fade of 20%.[70] Battery power rating was based on a four-hour discharge rate—*i.e.* 1500 MWh of capacity gives 375 MW of power capacity.

Most of the battery LCA studies relate to traction batteries for vehicles. Since stationary batteries often share the same cells as traction batteries, LCA results for cells across studies can be interchanged. Many of the stationary battery studies are focused on the consequential impact on grids during the LCA use phase.[69,71] Many of these studies use economic dispatch models on existing grids, which seek to maximise economic value. For this study, batteries do not interact with the grid, and therefore grid impacts of battery cycling are not applicable.

There is wide variation in LCA for Li-ion batteries, and few studies rely on primary data.[72] Ellingsen *et al.*[73] identified several issues in LCA studies of Li-ion batteries, including major differences in energy demand of cell manufacture. Life cycle GWP and cumulative energy demand for cells were taken from a review by Ellingsen *et al.*,[73] which was based on primary data from Ellingsen *et al.*[74] and Kim *et al.*[75] LCI data for housing, battery management system (BMS), inverter, cooling system, and insulation was taken from Pellow *et al.*[69] Minerals requirements are from Olivetti *et al.*[76] A functional unit of per kWh of storage capacity basis was used for cells, housings, BMS, and fire suppression system. A functional unit of per kW of battery power capacity was used for inverters and cooling systems. Li-ion batteries exhibit several degradation modes, including cycling, depth-of-discharge, and years of operation. For simplicity, an operational lifetime of 10 years was assumed.

2.5.3 Electrolyser plant. The electrolyser system is pressurised alkaline (30 bar) and based on a system investigated by Koj *et al.*[77] Each module is rated at 3.9 MW, 68 kg H_2 h^{-1}. Each electrolyser unit comprises 139-cell stack, gas separators (for separating O_2 and H_2 from electrolyte), electrolyte tanks for potassium hydroxide (KOH), KOH filter, heat exchangers (for cooling of O_2, H_2, and KOH), pumps, and power electronics. The baseline efficiency is set to 55 kWh kg^{-1} H_2 and sensitivity is conducted for 50 and 60 kWh kg^{-1} H_2. The full specifications are given in ESI,† Table S5.

At the cell level of the stack, pressurisation requires slightly more energy due to a higher overpotential than unpressurised electrolysis. The higher overpotential may be balanced against several offsetting properties. Gas bubble diameter is reduced at higher pressure reducing ohmic drop between electrodes; solubility is increased; and subsequent compression requirements are reduced.[78] Pressurised systems require thicker walled vessels with a commensurate increase in steel requirements, but a decrease in subsequent compressor requirements. The trade-off between atmospheric and pressurised systems is not assessed.

A small amount of KOH may be lost over time due to leakage or losses.[61] It is assumed that the KOH is replaced after 10 years. For maintenance, annual materials use of 1.5% of the construction based on an annual materials OPEX of 1.5% of CAPEX from[79] is assumed. One complete stack replacement is assumed over the operating life of the electrolyser. Since there are more than 100 stacks, maintenance and refurbishment would probably occur on a rotating basis at a prescribed time, or based on a measured performance decrease.

2.5.4 Compression and storage. The hydrogen is compressed further and stored onsite in an array of pressure vessels. The selected pressure depends on several factors, especially the required delivery pressure and mode of distribution. A pressure of 80 bar was selected as being suitable for pipeline distribution, liquefaction, or hydrogen feed for green ammonia production.[5,80]

For the baseline scenarios, 2 days of storage requires 380 cylindrical vessels, each 82 m^3 and 527 kg H_2 capacity. The material is 304 stainless steel, which has been shown to be largely immune to hydrogen effects at ambient temperature.[81] Each vessel is 2.4 m diameter, 20 metres long and 100 mm wall thickness (see ESI,† Table S6). Since the mass of pressure vessels scales roughly linearly with their storage capacity (additional wall thickness counters geometric scaling), the specific size of vessels is not critical to the LCA. Using the equation for ideal isentropic compression,[82] the specific work for post-electrolysis compression calculates to 0.42 kWh kg^{-1} H_2. Assuming a compressor isentropic efficiency of 0.8 gives 0.53 kWh kg^{-1} H_2, which is about 1% of hydrogen production energy use. LCI data for compressors is taken from ecoinvent 3.5.

2.5.5 Transmission system. The transmission system comprises transmission towers, insulators, conductors, and transformers. A 275 kV AC system was assumed as being representative for the power flows and distances traversed for up to 100 km, with voltage increased to 330 kV for 300 km. The specifications are given in ESI,† Table S7.

Transmission losses arise in line conductors, transformers, and switchgear. The most significant embodied energy component of the transmission system is the aluminium portion of the steel-aluminium conductors. This gives rise to a trade-off between conductor area and ohmic losses – a larger conductor area reduces ohmic losses (and therefore reduces required solar capacity for a given hydrogen production target), but increases the embodied energy and emissions of the transmission system. Conductor area was estimated using heuristics and considering resistive losses only. The baseline transmission distance was 20 km, and a sensitivity was conducted for 100 km and 300 km (see ESI,† Table S8). Transformer losses were assumed to be a constant 2%.[83]

LCI data for transformers was taken from Jorge *et al.*[84] A reference transformer of capacity 250 MVA from was selected and pro-rated to match the required transmission capacity, then scaled by 1.25 to provide capacity headroom. Equivalent transformers are required at each end of the transmission line. LCI data for transmission line foundations, masts, insulators, and conductors was taken from Jorge *et al.*[85] Preparation of the easement and roads was not included because of a lack of data,

Table 3 Headline results for baseline scenarios

Scenario	EROI	EPBT (years)	Emission intensity (kg CO_2 e kg^{-1} H_2)
Solar-battery (1500 MWh, 375 MW battery)	4.6	4.1	2.3
Solar-grid (grid imports 131 GW h $year^{-1}$: 7% of electrolysis energy)	5.0	3.7	4.3
Reference technology – steam methane reforming of natural gas[6,8]			8.9 (direct) to 11.9 (life cycle)

and that these are assumed to comprise a minor share of life cycle impacts.

2.5.6 Water system. H_2 demand of 100 tonnes per day corresponds to an annual feedwater demand of 329 ML $year^{-1}$. Reverse osmosis desalination using seawater was assumed as representative of a potable water solution in an arid, coastal location (see ESI,† Table S9). Desalination efficiency is assumed to be 3.8 kWh m^{-3} of water, which is about 0.1% of hydrogen production energy use. It is assumed that brine can be discharged to the sea in an environmentally sensitive manner.

2.5.7 Hydrogen production site. The hydrogen production site comprises paving, fences and buildings. It is assumed that the site would be located in an established industrial zone with road and service access. The total site area and building areas was estimated using a heuristic approach by comparison with comparable industrial plants where applicable, and the use of online mapping tools. It is assumed that the entire site would be paved with 200 mm, 40 MPa reinforced concrete using estimates from CCAA.[86]

3. Results

3.1 Overview of results

The most important results are summarised below, with more detailed discussion of results and sensitivities following.

(1) For both baseline scenarios, the GHG emission intensity is around a quarter that of H_2 produced from SMR.

(2) For both baseline scenarios, the EROI is less than comparable estimates for fossil fuels, and under some conditions, much less.

(3) For the solar-battery scenario, the solar modules were the most significant component that influenced EROI and GHG emissions.

(4) For the solar-grid scenario, the solar modules were important for EROI, but grid emissions were equally important for GHG emissions.

(5) Electrolyser turndown is a key sensitivity. The baseline turndown was set at 95%, but at 80 to 90% turndown, the EROI and GHG are adversely impacted. For the solar-battery configuration, a lower turndown (higher minimum electrolyser load) may be infeasible due to impractically large battery capacity.

(6) The emissions factor of electricity for the solar module supply chain is a critical sensitivity. Production in a lower emission region would significantly decrease the GHG emission intensity.

(7) Operational lifetime is important for both EROI and GHG intensity. Early decommissioning due to obsolescence, accelerated degradation, or premature failure would worsen both metrics.

3.2 Baseline scenarios

The headline results for the baseline scenarios are given in Table 3. Fig. 3 shows net-energy balance, with energy investments on a lifetime basis, and H_2 production on an annual basis for clarity. The baseline EROI is 4.6 for the solar-battery scenario and 5.0 for the solar-grid scenario. These estimates are somewhat lower than the global, final-stage EROI estimates of 6.0[15] to 8.5,[16] and therefore imply greater energy expenditures than incumbent fossil fuels. We return to this in Section 3.6.

The solar modules are the most significant component, with solar balance-of-system (frames, hardware, *etc.*) the second most significant. Interestingly, the electrolyser is much less significant than the solar modules despite a narrower difference in terms of CAPEX per unit of capacity. We return to this in Section 3.3.

Water demand for electrolysis is significant, at around 9 litres deionised water per kilogram hydrogen. However, the desalination plant is a minor component, both in terms of embodied energy and operating energy. Since arid regions tend to have both the highest solar radiation and lowest rainfall, siting of plants on or near a coast would seem sensible.

Fig. 4 shows unit GHG emissions for the solar-battery baseline scenario, and Fig. 5 shows unit GHG emissions for the solar-grid baseline scenario, both expressed as kg CO_2 e kg^{-1} H_2. The solar modules are significant for GHG results, but grid imports are also significant for the solar-grid scenario. At higher turndown, emissions due to grid imports are potentially much greater. We discuss this sensitivity in Section 3.4.

Batteries are not a major factor when modest storage is required. For the solar-battery scenario, a configuration was

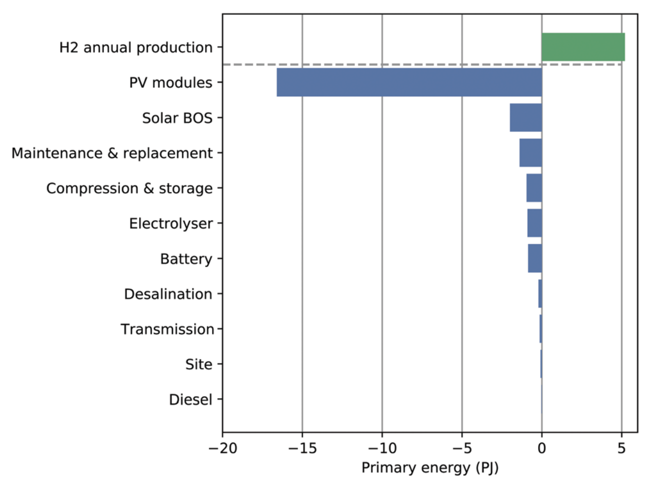

Fig. 3 Life cycle energy balance for solar-battery baseline scenario. H_2 production shown as annual flows.

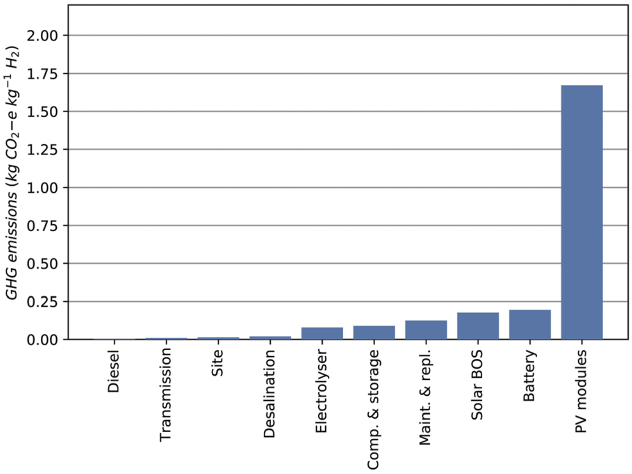

Fig. 4 Unit GHG emissions for solar-battery baseline scenario.

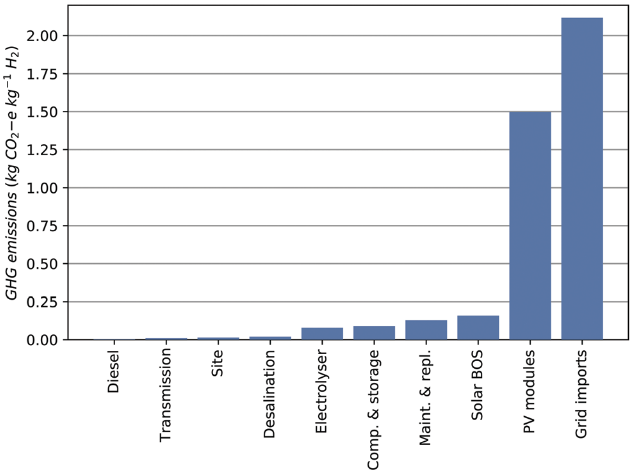

Fig. 5 Unit GHG emissions for solar-grid baseline scenario.

selected that traded-off slightly higher solar capacity *versus* lower battery capacity. At higher turndown, the required battery capacity increases significantly, which we return to in Section 3.5.

3.3 Comparison of solar and electrolyser system components

For both scenarios, the solar components are much more significant than the electrolysis components. A comparison of the materials load for the respective system components elucidates the underlying reasons for the difference.

In general, energy conversion devices that are lighter embody less materials and energy than functionally equivalent heavier devices, and depending on the materials and manufacturing processes used, will contribute less impacts than an equivalent heavier device. Since mass is inversely related to specific power, it follows that comparing the specific power of devices can provide guidance to their relative environmental impacts. The overall environmental impact needs to be weighed against the type of materials, manufacturing processes, and the raw materials need to produce the device. In the case of solar PV, an important factor is that of the requirement for very high purity silicon (solar grade silicon is defined as 99.9999% pure), although the processing energy has lessened in recent years due to more direct production pathways.[87]

A comparison between the solar farm and the electrolyser system is illustrative here because of the large differences in site area and specific power. The site area of the solar farm is 20.2 million m^2 compared to 0.033 million m^2 for the electrolyser building, or a roughly 600-fold difference; and the specific electrical power, averaged over a full year, of a solar module with mounting hardware is 2.0 watts kg^{-1} compared to 35.9 watts kg^{-1} for the electrolyser, or an 18-fold difference (see ESI,† Table S11). Hence the much larger site area for the solar farm, and much lower specific power supports the finding that the solar modules and associated hardware are much more critical to the impacts and embodied energy than the electrolyser system.

It is interesting that these differences are not replicated in the price difference per unit capacity between solar farms and electrolyser systems, whereby solar infrastructure is much less expensive than might be expected (or electrolysers costlier). IEA[2] reports a CAPEX of USD (2019) 1400 kW^{-1} for alkaline electrolysers, and IEA[88] reports a CAPEX of USD (2020) 800 to 2000 for utility scale solar PV. There are several factors that may help explain why there is a large difference between specific power and unit price. Bloomberg NEF[89] report a significant projected price difference between Western and Chinese-made electrolyser systems, and also a large differential between small ($<$50 kW) and large (2–3 MW) systems. Hence larger systems and greater Chinese manufacturing share should contribute to an electrolyser price decline in the future. On the other hand, a shift to Chinese manufacturing may worsen LCA/NEA metrics due to the comparatively higher GHG intensity of electricity in China, compared to the EU or USA for example.[57] Solar has undergone significant production learning, and commensurate cost decline, with IEA[90] reporting 716 GW of cumulative solar PV capacity in 2020. Bloomberg NEF[89] reported 138 MW of electrolysers built in 2018, hence there is probably less than 0.5 GW of installed capacity. The much lower electrolyser installed capacity suggests that electrolyser costs will decline significantly as cumulative production increases.

3.4 Emissions due to grid imports

Grid imports were the most significant contributor to the emissions embodied in the hydrogen in the solar-grid scenario despite comprising only 7% of energy used in the baseline grid scenario, with the embodied emissions of the solar modules also significant. For grid emissions, only the national reporting standard scope 2 (grid) emissions[56] were included. These comprise combustion and fugitive emissions, and comprise most of the life cycle emissions.[91]

The emissions arising from grid-electrolysis are given in Fig. 6 above, assuming 100% grid-electrolysis. This figure includes only those emissions arising from the grid (Scope 2), however these comprise most of the emissions except at very

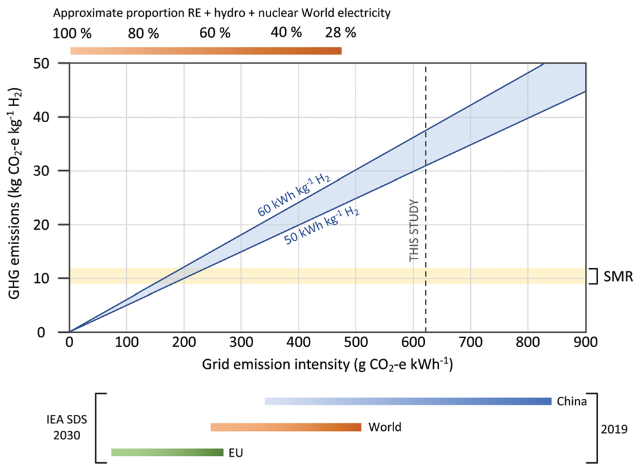

Fig. 6 Scope 2 greenhouse intensity per kg H_2 produced assuming 100% grid-electrolysis, but does not include full LCA emissions due to construction. The highlighted region shows the comparative emissions from SMR. GHG emissions due to electricity comprise only 2–3% of SMR lifecycle emissions.[8] Blue lines show emissions for electrolyser efficiencies of 50 & 60 kWh kg^{-1} H_2. The approximate proportion low emission electricity shown at the top is based on data from BP.[17] The ranges at the bottom are shown for 2019, and for 2030 using IEA Sustainable Development Scenario (SDS) ref. 52, 54 and 57.

low grid emissions factor ($<$200 g CO_2 e kWh^{-1}). As can be seen, the grid emission intensity must be very low to match SMR, and the proportion of electricity from RE (shown on the top axis) must be greater than around 70% before water electrolysis will improve on SMR.

3.5 Sensitivities

The GHG sensitivities for the solar-battery scenario are shown in Fig. 7 and the EROI sensitivities are shown in Fig. 8. Except for electrolyser turndown, the sensitivities for the solar-grid scenario are approximately the same. In this section, we discuss the sensitivities for both scenarios in more detail.

Electrolyser turndown. Ideally, electrolysers would have the capability of load-following solar generation at all times, and shutdown safely at night without control issues or accelerated stack degradation. In practice, all electrolysers are ramp-rate, turndown, and start-up constrained, with AE generally less flexible than PEM. To assess the sensitivity to turndown, it was necessary to re-run the production model (Section 2.4) to ascertain the required battery capacity and grid imports. Two turndowns were tested—80% and 90%. Future integrated plants may exhibit greater flexibility in load-following solar generation.

For the solar-grid scenario, the impact of reducing turndown (increasing minimum electrolyser load) is an almost linear increase in grid imports. These increased grid imports result in a higher GHG intensity—5.6 kg CO_2 e kg^{-1} H_2 at 90% turndown, and 8.8 kg CO_2 e kg^{-1} H_2 at 80% turndown.

For the solar-battery scenario, battery storage increase is highly non-linear at reduced turndown. There appears to be an inflection point at which storage requirements resolve to impractically large (several days of full-load storage equivalent) requirements. Storage requirements are determined by extended cloudy periods, during which there is insufficient daytime surplus solar to charge storage. These results suggest that an off-grid solar-battery system is feasible only at high ($>$90%) turndown. At 90% turndown, the plant EROI is 3.2 and GHG intensity 4.5 kg CO_2 e kg^{-1} H_2. At 80% turndown, there is a sharp upward inflection in battery storage requirements, such that the storage requirements cannot be stated with confidence, and therefore we do not provide EROI and GHG intensity at 80% turndown.

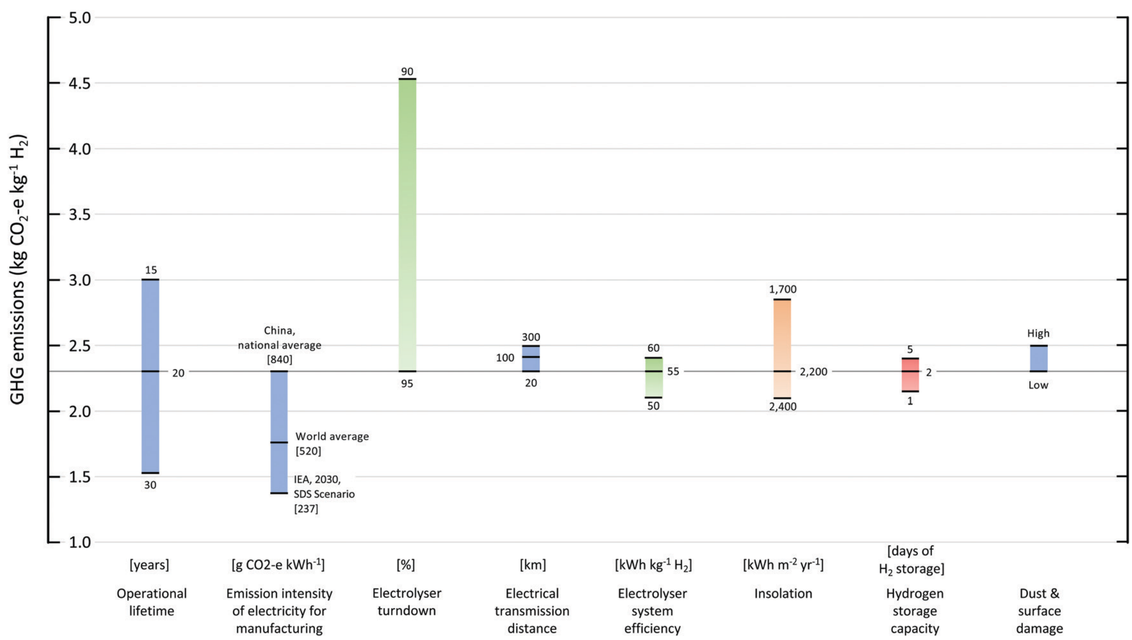

Fig. 7 GHG sensitivities for solar-battery scenario with baseline of 2.3 kg CO_2 e kg^{-1} H_2. Baseline is depicted by dark grey line.

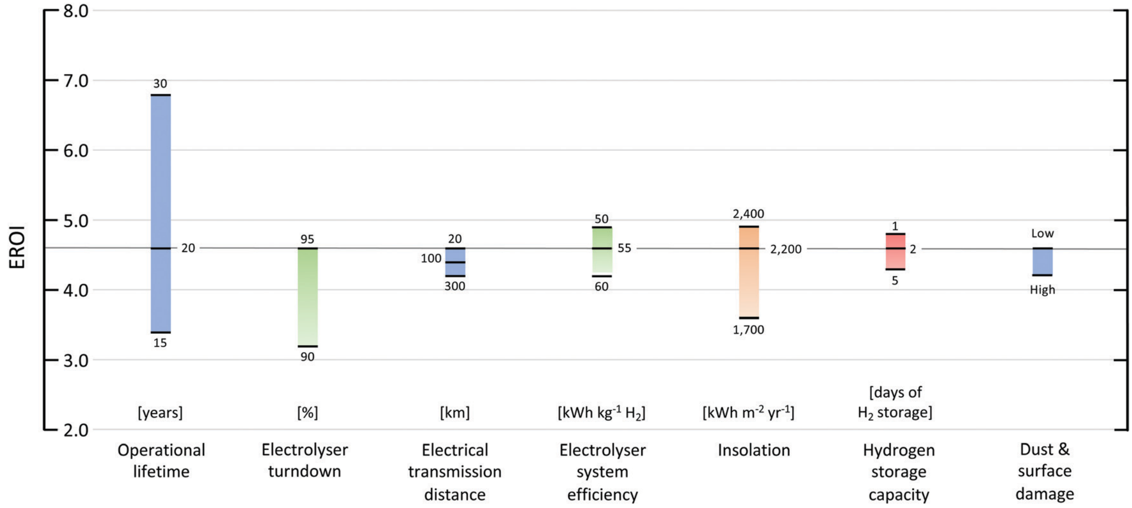

Fig. 8 EROI sensitivities for solar-battery scenario with baseline EROI of 4.6. Baseline is depicted by dark grey line.

Hydrogen storage capacity. Pressurised hydrogen storage makes more efficient use of manufacturing energy and materials compared to batteries, when expressed as the ratio of energy stored over a lifetime and the energy required to manufacture,.[92] To quantify the EROI and GHG difference between hydrogen storage and battery storage, a sensitivity analysis was conducted to assess the impact of providing an additional 3 days of storage. For this sensitivity, only the solar-battery scenario is examined. The baseline scenario included two days (200 tonne H_2) of hydrogen storage in steel pressure vessels, and batteries were used to power standby operation and maintain minimum electrolyser operation during low solar conditions. Two options for additional storage were considered. The first option was to assume increased pressure vessel total capacity to enable an additional 3 days of hydrogen storage, bringing total storage up to 5 days. This option increases GHG emission intensity by 6% and decreases EROI by 6%. The second option was to assume increased battery storage capacity equivalent to an additional 3 days of hydrogen storage by enabling electrolyser operation during low insolation conditions. This option increases GHG emission intensity by 79% and decreases EROI by 28%. Hence, it is preferable to increase storage capacity through enlarged pressurised hydrogen storage rather than enlarged battery capacity.

The EROI and GHG results reported here are in accord with studies that assess the cost difference between pressurised gas storage and battery storage. One such study used a least-cost optimisation of solar-electrolysis with battery storage and found that pressurised hydrogen storage was an order of magnitude less expensive than battery storage.[33]

Electrical transmission distance. Land requirements for the solar farm are of the order of 20 km^2 per gigawatt installed solar capacity. Competing uses of land, as well as cultural, financial, and social considerations, may preclude optimal siting of solar plants close to the coast, ports, flat terrain, or industrial areas. The baseline distance was set to 20 km, with sensitivity for 100 km and 300 km to accommodate siting in a location remote from the electrolysis facility.

Transmission distance affects the LCA and NEA results in two ways. The increased distance increases electrical resistive losses, requiring greater solar capacity to meet a given hydrogen production target. Secondly, the material mass of conductors and towers increases significantly over longer distances, thereby increasing embodied energy and emissions. Despite the significant materials involved in extending and upgrading transmission, this sensitivity does not significantly alter the overall results.

Electrolyser efficiency. Electrolyser efficiency is a key specification for electrolysis powered by solar PV. The efficiency determines the solar and transmission capacity required to meet a given target hydrogen production. The baseline efficiency of 55 kWh kg^{-1} H_2 is representative of the benchmark efficiency for megawatt scale AE.[89] The practical lower bound is around 45 kWh per kg H_2.[60] Sensitivity was conducted for 50 and 60 kWh kg^{-1} H_2.

From an LCA and NEA perspective, the efficiency of the electrolyser system is more significant than the energy and materials embodied in its construction. Hence, a more efficient electrolyser may be preferable even if it is somewhat heavier or more energy intensive to manufacture.

Insolation. The baseline GHI was 2200 kWh m^{-2} $year^{-1}$. Over the period 2000 to 2019, annual insolation at the modelled location (Learmonth, Western Australia) varied by +3/−5%. Average daily GHI varies from 4.0 kWh m^{-2} day^{-1} in June up to 8.3 kWh m^{-2} day^{-1} in December. The annual variation masks substantial monthly variance from year to year. For example, GHI during July has been as much as 23% below the 20 year average. Sensitivity was conducted for 1700 and 2400 kWh m^{-2} $year^{-1}$, reflecting approximate lower and upper annual estimates that are typical of the regions identified in this study.

Operational lifetime. Since the EROI and GHG intensity metrics are evaluated with respect to lifetime hydrogen production, the impact of a shorter, or longer, operational lifetime from baseline is a roughly linear worsening, or improvement, of those metrics. Early decommissioning due to obsolescence, accelerated degradation, or premature failure would worsen both metrics. On the other hand, optimised plant operation that minimises component degradation, and the use of advanced reliability techniques and monitoring may facilitate asset life extension, and therefore improve both metrics.

Solar panel dust and surface damage. Solar capture depends on the transparency of solar glass and is reduced by panel soiling and surface damage. Depending on the type of soiling (dust, dirt, bird droppings, leaves, *etc.*) rain will remove most of the soiling, with some residue remaining.[66] It is common for solar capture to be reduced by 10% due to soiling before rain. Annual, or more frequent, cleaning is usually carried out on large solar farms.[67] Additionally, installation in arid locations may result in surface damage to the hardened solar glass caused by windblown sand. There are several studies that have investigated sand damage, but a literature review did not reveal a reliable quantification of output decline. In a laboratory experiment replicating Saharan sandstorms, Bouaouadja *et al.*[93] found a 9% decline in light transmission after 5 hours of sandblasting. For this sensitivity, it is assumed that dust and surface damage reduces solar capture by 10% over an operational lifetime.

Combined sensitivities. The sensitivities just discussed and shown in Fig. 7 and 8 were estimated by changing one sensitivity while leaving all other sensitivities at their baseline values. Given a condition of multiple sensitivities that differ from baseline, the impact on EROI and GHG emissions is non-linearly additive. Taking all sensitivities for the solar-battery scenario together gives a minimum EROI (at minima for all sensitivities and 90% turndown) of 1.7 with a GHG intensity of 7.8 kg CO_2 e kg H_2; and a maximum EROI (maxima for all sensitivities and 95% turndown) of 8.1 with a GHG intensity of 0.8 kg CO_2 e kg H_2. Taking all sensitivities for the solar-grid scenario, gives a minimum EROI (for minima for all sensitivities and 80% turndown) of 2.6 with a GHG intensity of 10.2 kg CO_2 e kg H_2; and a maximum EROI (maxima for all sensitivities and turndown of 95%) of 8.9 with a GHG intensity of 2.7 kg CO_2 e kg H_2.

3.6 Scale of solar-electrolysis deployment

In light of the sheer scale of the prospective hydrogen demand, several challenges merit close consideration. In order to address one of these challenges, the mineral requirements to construct solar-electrolysis plants to meet prospective hydrogen demand in 2050 are evaluated.

There is a remarkably diverse range of estimates for hydrogen demand in 2050, ranging from almost no role at all, up to 30% of global final energy.[20] In the context of scenario analysis, it is important to distinguish between plausible and preferred futures,[94] and the deep uncertainties in future energy demand and substitution possibilities.[1] In the global integrated assessment model (IAM) field, the implementation of hydrogen is an emerging research area, however, the IAM literature to date suggests that the share of hydrogen is higher in strong abatement scenarios.[95] For this work, we adopt the REmap scenario from IRENA,[18] for which renewable hydrogen production in 2050 is 19 EJ year^{-1}, equivalent to 5% of total final energy consumption. It is assumed that half of IRENA's potential demand is met by solar-electrolysis, or 9.5 EJ year^{-1}, while noting that renewable hydrogen projections beyond 2050 are much greater.[20]

To meet this demand would require around 1800 of the solar-electrolysis plants with battery buffering considered in this study, operating on the baseline insolation of 2200 kWh m^2 year^{-1}. For this exercise, it is assumed that—(1) all of the plants constructed between now and 2050 will be operational in 2050; and (2) they will all embody current technology. Of course, future plants will embody the technologies and processes applicable at the time of construction. Furthermore, mineral supply is not static and driven by demand and price dynamics.

Six important minerals are shown in Fig. 9. Taking silver as an example, the construction of 1800 plants will require 0.24 Mt of silver, or 9 years of current annual silver production; for cobalt it is 7 years; for lithium it is 4 years; and so on. Other materials which are significant in absolute terms but less important in relative terms include steel at 124 Mt and concrete at 120 Mt. Three additional minerals used in PEM electrolysers are shown in Fig. 10, assuming that PEM electrolysers are adopted instead of alkaline electrolysers. PEM electrolyser stacks use platinum in the cathode, iridium in the anode,

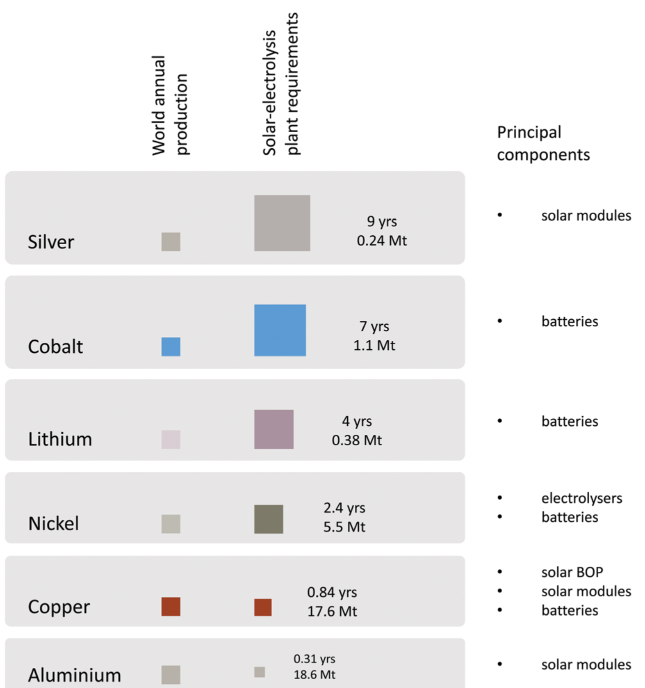

Fig. 9 Mineral requirements to construct solar-electrolysis plants to meet half of IRENA's 2050 hydrogen production target, expressed as years of production in 2018 and mineral requirements in million tonnes. The principal plant components show on right hand side. Mineral production is taken from USGS.[100]

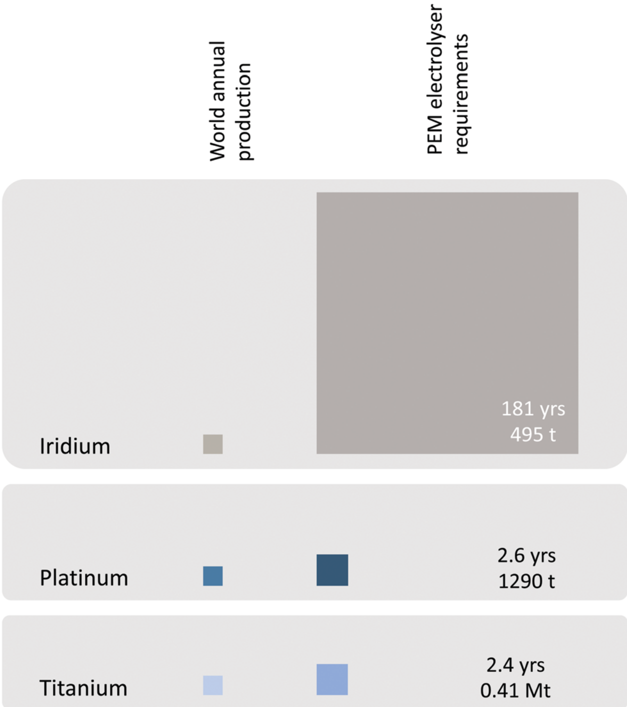

Fig. 10 Minerals requirements to construct PEM electrolyser stacks to meet half of IRENA's 2050 hydrogen production target (instead of alkaline electrolysers), expressed as years of production in 2016. Unit requirements and mineral production is taken from Kiemel *et al.*[31]

and titanium in the bipolar plates that separate anode and cathode. The platinum group metals are one of several critical materials for which no ready substitutes exist,[22] although substantial reductions in per unit resource use may be possible.[31]

Of the materials shown in Fig. 9 and 10, silver, cobalt, nickel, copper, aluminium, platinum, and titanium exhibit end-of-life recycling rates (EOL-RR) above 50%; iridium is 20–30%; and lithium is $<1\%$.[96,97] Because of increasing metal demand and long in-use lifetimes, recycled content of metal flows will remain significantly lower than EOL recycling rates for the foreseeable future.[96]

Since energy transition will involve the synchronous establishment of multiple supply-use pathways, there will be competing uses for minerals for construction of renewable electricity plants, batteries, transport and distribution infrastructure, end-use energy devices, along with the broad suite of industrial applications. One example of potential competing use is that between solar-electrolysis plants and battery electric vehicles. Cumulative battery capacity of the 1800 plants considered here is roughly equivalent to the total battery capacity of around 61 million electric vehicles, assuming battery capacity per vehicle of 45 kWh. This represents around 8-times the stock of battery electric vehicles (BEV) and plug-in hybrid electric (PHEV) vehicles in 2019.[98] Given that a shift of the global passenger vehicle fleet of roughly one billion vehicles to battery-electric will require a stock of minerals commensurate with, or greater than, currently available reserves for several critical minerals,[27] the implications of scaling across the entire decarbonisation effort need to be considered, and limits may impede energy transition.

A comparison of material flows for the construction of solar-electrolysis with natural gas flows for feed and fuel for SMR-based hydrogen production clarifies the headline results. To produce an equivalent supply of hydrogen to the 1800 solar-electrolysis plants discussed here would require an estimated 230 Mt year^{-1} natural gas using SMR, noting that the construction materials for SMR plant are much less than for solar-electrolysis. This represents around 64% of current global annual LNG demand of 359 Mt year^{-1}.[99] The substitution of SMR fossil fuel flows (and associated GHG emissions), with mineral flows for construction of solar-electrolysis plants, helps to explain the reduced GHG emissions of solar-electrolysis as well as the increased embodied energy (and therefore lower EROI).

The residual GHG emissions arising from construction of solar-electrolysis plants are due to fossil fuels that are embodied in global supply chains. Plants that are constructed in the future will embody the GHG emissions applicable at the time of construction, and are expected to decline with decarbonisation of global energy systems.

On the other hand, energy that is embodied in supply chains can be influenced by multiple factors, some of which will improve EROI, and some of which will worsen EROI. For example, technological progress that leads to reduced component mass or improved efficiency will tend to decrease embodied energy per unit of hydrogen produced and therefore improve EROI. Conversely, natural resource constraints and the commensurate need to exploit lower grade mineral resources will tend to increase embodied energy, and therefore worsen EROI. Similarly, siting of solar plants in lower insolation regions, sub-optimal plant design, or sub-optimal operation may reduce hydrogen production, and therefore lower EROI for a given capital investment.

Based on the results presented here, EROI is lower than the reference estimate for the global system comprising mostly fossil fuels, implying that a greater proportion of the overall economy's energy supply would need to be directed toward building renewable and hydrogen production infrastructure. Since nearly all the emissions associated with fossil fuel use are unmitigated, a fairer comparison would be to assess the EROI of fossil fuels with carbon capture and storage, or otherwise subsequent amelioration of impacts. Since mitigation efforts invariably require investment in additional infrastructure, and usually less energy delivered per unit of primary fuel, EROI of fossil fuels with mitigation would be expected to be lower than the reference EROI of fossil fuels without mitigation.

3.7 Limitations and further work

The specifications for the hypothetical facility were determined by heuristics and modelling. The objective of modelling was to establish realistic estimates for solar, electrolyser, and battery storage capacities for a hypothetical integrated plant.

Design for a prospective solar-electrolysis plant would include detailed site analysis and least-cost optimisation to guide decision making. Furthermore, commercial developers would be guided by other considerations not considered here. The specific product, market, and supply terms will be important for determining plant specifications. The alignment, or misalignment, of seasonal energy supply with seasonal energy demand may be important. An example of alignment might be southern hemisphere summer supply and northern hemisphere winter heating demand. Depending on location, a strategy of technology diversity (*e.g.* solar, wind, biomass), geographic diversity, and possibly product diversity (*e.g.* hydrogen, methanol) may be more cost-effective than dedicated water electrolysis powered by solar PV. Sector coupling *via* integration with electricity and natural gas networks, and possibly carbon capture and utilisation (CCU) for chemical production is another option. Hence, there would be differences between the configuration adopted for this study and the configuration that may be adopted for a commercial facility.

We assessed several important sensitivities, but there are other factors that could be considered in more detail. For example—(1) an average solar cell efficiency was used but the insolation-efficiency curve for crystalline silicon cells follows an inverted *U*-curve with minimum efficiency at low insolation;[101] (2) an average electrolyser efficiency was used but the efficiency curve for alkaline electrolysers is typically highest at low to medium load;[61] and (3) an average static solar tracking factor was used but the solar tracking factor varies with the annual cycle and the respective proportion of direct *versus* diffuse sunlight.[102] Further work could consider whether the additional accuracy from these factors justifies the additional complexity, or whether heuristic rules could be used to improve accuracy, but without undue complexity or specificity.

Battery storage was selected given the dynamic performance of Li-ion batteries, declining cost, and modularity. Modularity refers to small unit size and capability of connecting any number of units to achieve the required storage power and capacity. A suite of other storage options may be available that potentially offer reduced cost and lower impacts than battery storage, but that are constrained by geographical or other factors. Perhaps the leading example is pumped hydro storage (PHS), which is the dominant electrical storage technology on a globally installed capacity basis. No potential PHS sites within proximity to the modelled location in Western Australia Pilbara were identified, although one study that made use of topographical analysis identified potential off-river sites that might be worthy of further investigation.[103] Any such use would require additional analysis to assess environmental impacts.

An issue not included in this study is the treatment of carbon credits due to exports to the grid for the solar-grid scenario. From a physical flow perspective, flows from the grid embody the instantaneous emission intensity of grid electricity, and renewable flows to the grid reduce the instantaneous emission intensity of grid electricity. From a carbon accounting and regulatory perspective, a question arises as to the attribution of 'carbon credits' due to the reduction of the instantaneous emission intensity arising from renewable flows to the grid. This study was limited to a physical flow perspective, but clearly carbon accounting would be consequential from an economic perspective. The economics of carbon accounting was not considered for this study, but may be important for the economic viability of a plant, depending on the regulations applicable in the jurisdiction at the time.

There is considerable variation in the availability and quality of LCI data. Due to the proprietary nature of technical systems, firm-level primary data can be hard to obtain. Comprehensive data is available for solar modules and solar balance-of-plant. Comprehensive data on electrolysers is sparse. Most of the renewable-electrolysis LCA studies in the literature use secondary life cycle inventory data and there is little primary data available in the public domain. The primary data on electrolysis that is available seems to have come from groups that own/manage equipment rather than being sourced from OEMs. Similarly, most LCA studies examining batteries rely on secondary data or own estimates. Assumptions about technology learning rate given the development that may need to occur for such a plant to be established are a related factor. Given the upscaling of manufacturing capacity that will be required, there is considerable uncertainty of the future impacts along the full manufacturing supply chain.

4. Conclusions

Given the global decarbonisation challenge, and commensurate upscaling required for renewable hydrogen, it is critical that the environmental impacts and energetic performance of renewable hydrogen technologies are understood.

In this work, we undertook an assessment of a hypothetical large scale solar-electrolysis facility to investigate the factors that influence greenhouse gas (GHG) emissions and net energy performance. We sought to address a gap in the environmental assessment literature, being the insufficient attention given to context specific variances, operating constraints, and sensitivities of prospective large scale solar-electrolysis plants. Traditionally, life cycle assessment (LCA) studies are retrospective studies of existing technologies. The dilemma is that there are no large-scale renewable-electrolysis plants in operation from which to collect operating data and plant specifications. Instead, LCA practitioners generally make the simplifying assumptions of steady state operation under average conditions. Whilst these assumptions may be necessary for preliminary analysis, insufficient consideration is given to the marked differences arising from context specific variances and the physical and thermodynamic constraints of electrolysis and post-electrolysis processing. We assessed important factors that influence the environmental and energetic performance of solar-electrolysis and sought to identify other factors that could be considered in more detail. There is a critical need for this given that steady state analysis studies generally show low energy returns on the energy invested in the building of the plants.

Under baseline conditions, GHG emissions are around a quarter that of steam methane reforming (SMR). However, under reasonably anticipated conditions with grid buffering, the GHG emissions may be comparable to SMR. The most important sensitivity is electrolyser turndown, and the most important component is the solar module. Other important sensitivities include plant operating lifetime, the electricity emissions factor of the solar module supply chain, and strength and variability of solar resource. Assuming a declining GHG emission intensity of global energy, GHG emission intensity will also decline. However, hydrogen will embed the emissions profile of the specific region, or regions, of the supply chain at the time of manufacture.

We found that EROI was lower than the reference EROI for fossil fuels under baseline scenarios, and under some conditions with multiple sensitivities, much lower. Ongoing efficiency improvements in solar module manufacturing will drive improvements in EROI, while resource constraints will worsen EROI.

Further work needs to be undertaken to ascertain turndown and ramp rates of electrolysers and post-electrolysis processes at scale, the impact on control, safety, degradation and performance, and integration with hydrogen liquefaction or ammonia plants. Design for sustainability implies that life cycle parameters need to be treated as objective functions in plant optimisation. We recommend that LCA and NEA are integrated with project planning to inform decision making to ensure that hydrogen meets the goals of sustainable production.

Nomenclature & abbreviations

AC	Alternating current
AE	Alkaline electrolyser
BMS	Battery management system
CAPEX	Capital expenditure
CCS	Carbon capture and storage
CCU	Carbon capture and utilisation
EROI	Energy return on (energy) invested
EPBT	Energy payback time (years)
HHV	Higher heating value (joules)
HQ	High-cubed (container)
IEA	International Energy Agency
IEA-PVPS	IEA Photovoltaic Power Systems Programme
Li-ion	Lithium ion battery
KOH	Potassium hydroxide
GHG	Greenhouse gas
GHI	Global horizontal irradiance (kWh m^{-2} $year^{-1}$)
GWP	Global warming potential (kg CO_2 e)
ISO	International Organization for Standardization
LCA	Life cycle assessment (or analysis)
LCI	Life cycle inventory
NEA	Net-energy analysis
NMC	Nickel manganese cobalt
Nm^3	Normal cubic metres
NWIS	North West Interconnected System (Western Australia)
OPEX	Operational expenditure
PEC	Photoelectrochemical cells
PEM	Proton exchange membrane (electrolysers)
PPA	Power purchase agreement
PR	Performance ratio
PV	Photovoltaics
SMR	Steam methane reforming

Conflicts of interest

There are no conflicts of interest to declare.

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